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Debris-Extracting Hairbrush with Integrated Woven Mesh Filter for Gentle Cleaning

Abstract

The invention discloses a hairbrush and an attachment for a hairbrush, each designed to facilitate the removal of sand and debris from hair while minimizing potential damage. The hairbrush comprises a brush body, multiple bristles affixed thereto, and a woven nylon mesh layer positioned above the bristle base. This configuration allows only the tips of the bristles to extend through the woven nylon mesh by 1 to 4 millimeters, effectively capturing debris and allowing for gentle cleaning. The snap-on attachment mirrors this functionality with a layer of woven nylon mesh and securing mechanisms for attachment to a variety of brush bodies, enabling the bristle tips to protrude similarly for debris removal. Both the hairbrush and the attachment are designed to improve hair care routines by offering an efficient and gentle means of removing particles from hair, thereby addressing the need for a hair cleaning tool that balances effectiveness with the preservation of hair integrity.

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Background/Summary

FIELD OF INVENTION

[0001] The present invention relates generally to the field of personal grooming devices, specifically to a hairbrush designed for the removal of sand and debris from hair, incorporating a unique woven mesh filter system to ensure gentle yet effective cleaning.

BACKGROUND

[0002] In the realm of personal grooming and hair care, particularly in the context of removing foreign particles such as sand and debris from human hair, existing methods and tools have consistently fallen short of meeting the dual needs of effectiveness and gentleness.

[0003] Traditional approaches to this issue, while somewhat effective in removing unwanted particles, often lead to undesirable consequences such as hair damage, tangling, and discomfort during the cleaning process. This is largely due to the inherent design limitations of conventional hairbrushes, which do not adequately prevent the entanglement of debris within the bristles, nor do they cater to the delicate nature of human hair. The bristles of these brushes, often designed without consideration for the gentle removal of particles, frequently cause further entanglement and can damage both the hair and scalp.

[0004] Moreover, while there have been attempts to innovate within the domain of brush design, such as the introduction of brushes with removable one-time use nylon layers, these innovations have primarily focused on applications for grooming animals. These designs are not suited to the unique requirements of human hair care, as they tend to trap hair and debris in a manner that can be damaging to human hair. The brushes intended for animals often fail to address the need for a tool that combines effective debris removal with a gentle touch that is essential for human hair care.

[0005] The conventional methods and tools available have proven inadequate, as they do not effectively balance the need for thorough cleaning with the need for hair and scalp protection. The limitations and drawbacks of current systems underscore the necessity for a solution that can navigate the fine line between efficacy and gentleness.

[0006] It is within this context that the present invention is provided.

SUMMARY

[0007] The present invention relates to a hairbrush and a snap-on attachment for a hairbrush, each designed to facilitate the removal of sand and debris from hair while minimizing potential damage. The hairbrush comprises a brush body, multiple bristles affixed thereto, and at least one tightly woven nylon mesh layer positioned above the bristle base. This configuration allows only the tips of the bristles to extend through the nylon mesh by 1 to 4 millimeters, effectively capturing debris and allowing for gentle cleaning. The attachment mirrors this functionality with a layer of tightly woven nylon mesh and securing mechanisms for attachment to a variety of brush bodies, enabling the bristle tips to protrude similarly for debris removal. Both the hairbrush and the attachment are designed to improve hair care routines by offering an efficient and gentle means of removing particles from hair, thereby addressing the need for a hair cleaning tool that balances effectiveness with the preservation of hair integrity.

[0008] It is important to distinguish the tightly woven fabric of the nylon mesh layer of the present invention from prior art designs that incorporate wire grids and the like. The holes of the mesh layer should be of a size which prevents debris as small as sand from falling through. Only a tightly woven fabric such as nylon can achieve this, maintaining good structural integrity whilst still providing the required elasticity to poke the brush bristles through the same small gaps. For example, a nylon fabric with gaps of a width of 0.5 mm or less may be used for the mesh layer.

[0009] According to a first aspect, a hairbrush is provided that includes a brush body, a plurality of bristles affixed to the brush body, and a layer of woven nylon mesh positioned above the base of

the bristles. This configuration allows only the tips of the bristles to extend through the nylon mesh, and provides a tightly woven netting for effectively capturing debris while allowing the bristles to gently clean the hair.
[0010] In some embodiments, the bristles of the hairbrush are made of boar hair, which is known for its natural ability to distribute oils throughout the hair, contributing to a healthier scalp and hair.
[0011] Further embodiments include a rubber base located beneath the woven nylon mesh, providing support for the mesh and bristles, thereby enhancing the brush's overall durability and effectiveness in debris removal.
[0012] Additionally, some embodiments feature a layer of felt positioned between the rubber base and the woven nylon mesh. This felt layer enhances the brush's ability to capture debris, making the cleaning process more efficient.
[0013] The brush body, in certain embodiments, is constructed from materials such as plastic or wood, offering choices to consumers based on their preferences for durability, aesthetics, and environmental impact.
[0014] For increased hygiene and maintenance ease, the woven nylon mesh in certain embodiments is integrated into the brush as a non-removable component, ensuring that the mesh remains securely in place during use.
[0015] Designed for versatility, the hairbrush can be used in dry conditions or in conjunction with water or hair conditioner, facilitating the removal of sand and debris in various situations.
[0016] The nylon mesh is characterized by a tightly woven network-like structure with small, evenly spaced holes, optimized for trapping particles while allowing gentle passage of hair, reducing the risk of damage during the cleaning process.
[0017] An ergonomic handle is featured in some embodiments, molded to fit the contours of a user's hand, improving grip and control during use.
[0018] According to another aspect, the invention encompasses an attachment for hairbrushes, featuring a layer of woven nylon mesh and securing mechanisms for easy attachment to various brush bodies. This attachment allows for the bristles of an existing hairbrush to protrude through the nylon mesh, facilitating debris removal while minimizing hair damage.
[0019] Some embodiments of the attachment utilize clips as securing mechanisms, offering a simple and effective method for attaching the mesh to hairbrushes.
[0020] In other embodiments, Velcro straps serve as the securing mechanisms, allowing for adjustable tension and accommodation of hairbrushes of varying sizes and shapes, enhancing the attachment's versatility.
[0021] Further embodiments employ snap-fit connectors as securing mechanisms, providing a secure and easily detachable method of attaching the mesh to the hairbrush body, facilitating ease of use and adaptability to different brush designs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] Various embodiments of the invention are disclosed in the following detailed description and accompanying drawings.
[0023] FIG. **1** provides an isometric perspective view of the hairbrush, showcasing its overall structure including the handle, brush body, and the layers comprising the rubber base, felt layer, and nylon mesh positioned on the front pad for debris interception.
[0024] FIG. **2** displays a side view of the hairbrush, emphasizing the convex curvature of the rubber base, felt layer, and nylon mesh.
[0025] FIG. **3** illustrates a snap-fit attachment in a perspective view about to be placed over a standard soft hairbrush. This figure highlights the attachment's mesh layer and side supports

equipped with securing mechanisms, designed to retrofit existing hairbrushes with the debris removal features of the invention.

[0026] Common reference numerals are used throughout the figures and the detailed description to indicate like elements. One skilled in the art will readily recognize that the above figures are examples and that other architectures, modes of operation, orders of operation, and elements/functions can be provided and implemented without departing from the characteristics and features of the invention, as set forth in the claims.

DETAILED DESCRIPTION AND PREFERRED EMBODIMENT

[0027] The following is a detailed description of exemplary embodiments to illustrate the principles of the invention. The embodiments are provided to illustrate aspects of the invention, but the invention is not limited to any embodiment. The scope of the invention encompasses numerous alternatives, modifications and equivalent; it is limited only by the claims.

[0028] Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. However, the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

DEFINITIONS

[0029] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention.

[0030] As used herein, the term “and/or” includes any combinations of one or more of the associated listed items.

[0031] As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well as the singular forms, unless the context clearly indicates otherwise.

[0032] It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0033] As used herein, the term “brush body” refers to the main structure of a hairbrush to which bristles are attached. This structure may be made from a variety of materials, including but not limited to wood, plastic, metal, or composite materials, each selected for properties such as durability, weight, and aesthetic appeal.

[0034] The term “bristles” as described herein encompasses any hair-like projections used for the purpose of grooming, cleaning, or detangling hair. Bristles may be made from natural materials such as boar hair, synthetic materials like nylon, or a combination thereof. The choice of bristle material affects the brush's functionality, including its ability to distribute natural oils through the hair, remove debris, and minimize damage to the hair and scalp.

[0035] As used herein, “nylon mesh” refers to a tightly woven fabric or material characterized by its mesh-like structure, made from nylon or similar synthetic polymers, designed to filter or trap debris. The mesh's specifications, including hole size and distribution, are tailored to efficiently capture sand and debris while allowing hair to pass through with minimal resistance, thereby reducing the risk of tangling and damage.

[0036] The term “securing mechanisms” as described herein refers to devices or systems employed to attach or secure two components together. In the context of the present invention, securing mechanisms may include but are not limited to snap-fit connectors, clips, Velcro straps, magnetic fasteners, or any other suitable means that enable the detachable connection of the snap-on attachment to a hairbrush. These mechanisms are designed for ease of use, allowing for quick attachment and removal, and can be adapted to fit a wide range of hairbrush sizes and styles.

DESCRIPTION OF DRAWINGS

[0037] The present invention relates to a hairbrush and an attachment for a hairbrush, both designed to address the common problem of removing sand and debris from hair. The invention includes a brush body equipped with a plurality of bristles and a novel layering system that incorporates a nylon mesh. This mesh is strategically positioned to allow only the tips of the bristles to extend through, effectively capturing debris while minimizing damage and tangling of hair during the cleaning process. The design focuses on providing a solution that is both effective in debris removal and gentle on the hair and scalp, addressing limitations found in conventional hairbrush designs.

[0038] Additionally, the invention encompasses an attachment that can be applied to standard hairbrushes, extending the benefits of the debris-removal system to a wider range of existing grooming tools. This attachment features a similar nylon mesh layer and is equipped with securing mechanisms that facilitate easy attachment and removal, making it adaptable to various brush sizes and styles.

[0039] Referring to FIG. **1**, an isometric perspective view is shown of an example hairbrush **100** according to the present invention. The hairbrush **100** comprises a handle **102**, a brush body **104**, both of which are made of wood or plastic. The front pad of the hairbrush comprises a convex rubber base **106**, on top of which a felt layer **108** and tightly woven nylon mesh fabric **110** are positioned. The woven nylon fabric layer **110** rests on top of the felt **108**.

[0040] The holes of the woven nylon mesh **110** are not to scale, they are shown larger than they will actually be for clarity of illustration.

[0041] The tips of the brush bristles **112** extend through the gaps in the nylon mesh, with only a small portion, 1-4 mm (ideally 2-3 mm) exposed. These bristles may be made of a gentle bristle type such as boar hair. Debris such as sand that is drawn out of a person's hair by the brush will be caught by the nylon mesh **110** as the holes will be sized small enough that even sand cannot fall through, allowing captured debris to fall to the floor or be easily shaken out of the brush **100**. For example, a nylon fabric with gaps of a width of 0.5 mm or less may be used for the mesh layer. Any debris that does get through the woven mesh layer **110** is caught by the felt layer, and is also easily removed. Debris cannot reach the base of the bristles **112**, and the bristles cannot become entangled with each other.

[0042] Referring to FIG. **2**, a side view is shown of the same brush **100**. The convex curve formed by the layers is more apparent in this view.

[0043] Referring to FIG. **3**, a snap-fit attachment **200** of an alternative embodiment of the invention is shown about to be placed over a standard soft hairbrush **300** to provide it with similar qualities as the brush of the invention. The attachment comprises a tightly woven mesh layer **202** similar to that of the brush **100**, but which is held between a pair of side supports **204** that are provided with securing mechanisms such as snap fit attachments, clips, Velcro, or any other suitable mechanisms for securing to the sides of a brush so that the mesh layer rests over the bristles **302** with only 1-4 mm of bristle exposed.

CONCLUSION

[0044] Unless otherwise defined, all terms (including technical terms) used herein have the same meaning as commonly understood by one having ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0045] The disclosed embodiments are illustrative, not restrictive. While specific configurations of the hairbrush of the invention have been described in a specific manner referring to the illustrated embodiments, it is understood that the present invention can be applied to a wide variety of solutions which fit within the scope and spirit of the claims. There are many alternative ways of implementing the invention.

[0046] It is to be understood that the embodiments of the invention herein described are merely illustrative of the application of the principles of the invention. Reference herein to details of the illustrated embodiments is not intended to limit the scope of the claims, which themselves recite those features regarded as essential to the invention.

Claims

1. A hairbrush comprising: a brush body; a plurality of bristles affixed to said brush body, wherein each of said bristles extends from a base connected to said brush body; and a layer of woven nylon mesh positioned above said base of the bristles and attached to said brush body, wherein said woven nylon mesh is configured to allow only the tips of said bristles to protrude through said nylon mesh, said tips extending a length of between 1 millimeter and 4 millimeters beyond said nylon mesh, characterized in that the woven nylon mesh serves to intercept and retain debris while allowing said tips of the bristles to engage with and clean hair.
2. The hairbrush of claim 1, wherein the woven nylon mesh is a fabric with gaps of a width of 0.5 mm or less.
3. The hairbrush of claim 1, wherein said bristles are made of boar hair.
4. The hairbrush of claim 1, further comprising a rubber base located beneath said nylon mesh, wherein said rubber base provides support for the nylon mesh and bristles.
5. The hairbrush of claim 4, further comprising a layer of felt positioned between said rubber base and said nylon mesh to enhance debris capture capabilities.
6. The hairbrush of claim 1, wherein said brush body is constructed from a material selected from the group consisting of plastic and wood.
7. The hairbrush of claim 1, wherein said nylon mesh is integrated into the brush as a non-removable component.
8. The hairbrush of claim 1, designed for use in a dry condition or in conjunction with water or a hair conditioner to aid in the removal of sand and debris from hair.
9. The hairbrush of claim 1, wherein the nylon mesh comprises a network-like structure with evenly spaced holes or openings sized to trap particles of sand and debris while allowing hair to pass through gently.
10. The hairbrush of claim 1, wherein the bristles are configured with tapered tips to facilitate gentle engagement with the scalp and hair, reducing the risk of damage during debris removal.
11. The hairbrush of claim 1, further comprising an ergonomic handle designed to improve grip and control during use, said handle being moulded to fit the contours of a user's hand.
12. The hairbrush of claim 1, wherein the nylon mesh is coated with an antimicrobial agent to inhibit the growth of bacteria and mould, ensuring the brush remains hygienic over extended periods of use.
13. The hairbrush of claim 1, further comprising a mechanism for adjusting the tension of the nylon mesh, allowing for the accommodation of different hair types and debris sizes.
14. An attachment for a hairbrush, comprising: A layer of woven nylon mesh configured to overlay the bristles of a hairbrush; and a pair of securing mechanisms located on opposing sides of the attachment, wherein each of the securing mechanisms is adapted to removably attach the attachment to the sides of the hairbrush body, characterized in that the woven nylon mesh is positioned to allow only the tips of the bristles of the hairbrush to protrude through the woven nylon mesh, with said tips extending a length of between 1 millimeter and 4 millimeters beyond said nylon mesh, thereby enabling the mesh to intercept and retain debris while permitting the exposed tips of the bristles to engage with and clean hair.
15. The attachment of claim 14, wherein the woven nylon mesh is a fabric with gaps of a width of 0.5 mm or less.
16. The attachment of claim 14, wherein the securing mechanisms comprise clips designed to snap onto the existing structure of a hairbrush

17. The attachment of claim 14, wherein the securing mechanisms are Velcro straps designed to wrap around the hairbrush body, the Velcro straps allowing for adjustable tension and accommodation of hairbrushes of different sizes and shapes.
18. The attachment of claim 14, wherein the securing mechanisms are snap-fit connectors that engage with corresponding grooves or notches on the hairbrush body.